Huancheng Chen

huanchengch@gmail.com | citychan.github.io | huanchengch | CityChan

Education

The University of Texas at Austin, Austin, Texas 2020 – 2025

Ph.D. in Electrical and Computer Engineering

South China University of Technology, Guangzhou, China 2015 – 2019

B.E. in Electrical Engineering and Automation

Research Overview

My research interests lie in LLM post-training and efficient LLMs, with a focus on making large language models more capable, reliable, and deployable. Specifically, I am interested in:

- **LLM post-training:** SFT, RL to learn memory management and self-evolving skills. **[13, 14, 15, 16]**
- **Efficient LLMs:** inference acceleration via quantization, pruning, and distillation. **[3, 4, 9, 11]**
- **Continual learning:** fine-tuning pretrained LLMs and VLMs on downstream tasks. **[8, 9, 10, 12]**
- **Trustworthy AI:** privacy-preserving machine learning and model safety. **[2, 5]**

Industrial Experience

Microsoft AI, Mountain View, California Mar. 2026 – present

Applied Scientist II

- **Foundation model training (Tech Lead):**
  - Design sparse-attention, MoE, and LooP models at 300B, 30B, 4B, and 1B parameter scales.
  - Build fine-tuning/RL Infra (MFU 0.1%→5%, 10T data, 1,000 MI300X GPUs).
  - Build EvalHub for agentic benchmarking, unifying Docker, environments, data, and MCP.
- Pretrain and post-train multimodal retrieval models on PB-scale text, image, and video data for Bing.
- Contribute to the development of multiple products: Web IQ, Copilot AI Mode, and Copilot Overview.
- Lead research projects on RSI, agent skill evolution (SkillSpec), and hierarchical skill routing (TalentRL).

Accenture, Mountain View, California Jun. 2025 – Mar. 2026

Senior Research Scientist at Advanced AI Center

- Contributed to Accenture's research on memory management for long-horizon LLM agents via MemexRL.
- Contributed to Accenture's research on efficient RL for improving SLM's tool-use capabilities via CacheRL.
- Contributed to the Training-as-a-Service functionality (training, evaluation, and data curation), an end-to-end pipeline that enables clients to customize LLMs with their own data on the AI Refinery platform.

Sony AI, Austin, Texas & Tokyo, Japan Feb. 2024 – Aug. 2024

Research Intern at PPML Team

- Designed MAC, a zero-shot method that enhances spatial semantics in diffusion-based layout-to-image generation without additional training.
- Proposed DualLoRA, a novel PEFT method for continually fine-tuning pretrained ViTs on downstream tasks without data replay.

Toyota InfoTech Lab, Mountain View, California May. 2022 – Aug. 2022

Research Intern at Infrastructure & Data Platform

- Applied knowledge distillation in FedHKD to mitigate the model drift problem in non-IID distributed learning and protect local data privacy.

Bell Labs, Murray Hill, New Jersey Jan. 2022 – May. 2022

Research Intern at Mathematics & Algorithms Research Group

- Developed a U-2-Net-based framework for removing irrelevant background to improve flaw detection.

PUBLICATIONS

($\star$ indicates equal contribution)

[NeurIPS] [11] S. Cha, **H. Chen**, D. Kim, H. Zhang, K. Chan, G. Veciana, H. Vikalo, "CoreQ: Learning-Free Mismatch Correction and Successive Rounding for Quantization", *Advances in Neural Information Processing Systems*, 2026. **Spotlight (top 1%)**

[TMLR] [10] S. Cha*, **H. Chen***, H. Vikalo, "FedProTIP: Task-Agnostic Federated Continual Learning via Replay-Free Gradient Projection", *Transactions on Machine Learning Research*, 2026.

[ICLR] [9] D. Kim, S. Cha, **H. Chen**, C. Wang, H. Vikalo, "Quantized Gradient Projection for Memory-Efficient Continual Learning", *International Conference on Learning Representations*, 2026.

[CVPRW] [8] **H. Chen**, J. Li, W. Zhuang, C. Chen, L. Lyu, "Replay-Free Continual Low-Rank Adaptation with Dynamic Memory", *Computer Vision and Pattern Recognition Workshop*, 2026.

[CVPRW] [7] **H. Chen**, J. Li, W. Zhuang, H. Vikalo, L. Lyu, "Training-Free Layout-to-Image Generation with Marginal Attention Constraints", *Computer Vision and Pattern Recognition Workshop*, 2026.

[NeurIPS] [6] **H. Chen**, H. Vikalo, "Heterogeneity-Guided Client Sampling: Towards Fast and Efficient Non-IID Federated Learning", *Advances in Neural Information Processing Systems*, 2024.

[ICML] [5] **H. Chen**, H. Vikalo, "Recovering Labels from Local Updates in Federated Learning", *International Conference on Machine Learning*, 2024.

[CVPR] [4] **H. Chen**, H. Vikalo, "Mixed-Precision Quantization for Federated Learning on Resource-Constrained Heterogeneous Devices", *Computer Vision and Pattern Recognition*, 2024.

[ICLR] [3] **H. Chen**, C. Wang, H. Vikalo, "The Best of Both Worlds: Accurate Global and Personalized Models through Federated Learning with Data-Free Hyper-Knowledge Distillation", *International Conference on Learning Representations*, 2023.

[CVPRW] [2] **H. Chen**, H. Vikalo, "Federated Learning in Non-IID Settings Aided by Differentially Private Synthetic Data", *Computer Vision and Pattern Recognition Workshop*, 2023. **Oral**

[ICCVW] [1] A. Mohamed*, **H. Chen***, Z. Wang, C. Claudel, "Skeleton-Graph: Long-Term 3D Motion Prediction From 2D Observations Using Deep Spatio-Temporal Graph CNNs", *International Conference on Computer Vision Workshop*, 2021.

PREPRINTS

[16] **H. Chen**, X. Sun, Z. Huang, M. Chang, J. Lu, "TalentRL: Strategy-Guided Reinforcement Learning for Hierarchical Skill Routing", *Under Review*.

[15] **H. Chen**, X. Sun, Z. Huang, S. Huang, S. Singhal, J. Lu, "SkillSpec: Consensus-Gated Agent Skill Evolution via Representation Specialization", *Under Review*.

[14] M. Islam, S. Thakur, **H. Chen**, S. Park, J. Wang, G. Kim, "CacheRL: Multi-Turn Tool-Calling Agents via Cached Rollouts and Hybrid Reward", *Technical Report*.

[13] Z. Wang, **H. Chen**, J. Wang, W. Wei, "Memex(RL): Scaling Long-Horizon LLM Agents via Indexed Experience Memory", *Technical Report*.

[12] D. Kim, A. Wynter, **H. Chen**, H. Kim, H. Vikalo, "Foundation-Preserving Adaptation via Generalized Rayleigh-Quotient Optimization", *Under Review*.

SKILLS

- Platforms: Linux, Azure, AWS, GCP
- Programming: Python, MATLAB, Java, C/C++, LaTeX
- Agent & training stacks: PyTorch, verl, slime, SGLang, vLLM, Ray, SkyPilot, Kubernetes

COMMUNITY SERVICES

- Conference Reviewer: ICML (22,23,24,25,26), NeurIPS (22,23,24,25,26), ICLR (24,25,27), IJCAI (24,25), AAAI (25), CVPR (24,25), ACL (26).
- Journal Reviewer: IEEE Transactions on Mobile Computing